Literature Status for arXiv:1501.01685: The Riesz–Kantorovich Modulus Question

Run fa_banach_001, agent_lane_05

11 August 2026

Status

This is a `literature_already_answered` packet. Troitsky and Xanthos recall a classical open problem in their paper *Spaces of regular abstract martingales*, arXiv:1501.01685 [1]. Michael Elliott later answered that problem negatively in *The Riesz–Kantorovich formulae* [2].

Source Question

In Section 2, on page 5 of the locally rendered source PDF, Troitsky and Xanthos write that it is open whether the modulus of an operator is always given by the Riesz–Kantorovich formula. More explicitly, if a linear operator $T : X \to Y$ between vector lattices has a modulus $|T|$ in the ordered space of operators, must one have, for every $x \in X_+$,

$$|T|x = \sup T[-x, x]?$$

The source cites Aliprantis–Tourky for this problem and then turns to the analogous Krickeberg formula for abstract martingales.

Supporting Answer

Elliott's abstract says that the paper constructs an operator whose modulus does not satisfy the Riesz–Kantorovich formula, thereby answering the long-standing question. On the second journal page, the introduction states the main result explicitly:

> There exist a compact Hausdorff space K and a regular linear operator $R : L^1[0, 1] \to C(K)$ having a modulus which does not satisfy the Riesz–Kantorovich formula.

The same introduction identifies the long-standing question, cites Aliprantis–Tourky's monograph, and says that the theorem resolves it. Hence the general question recalled in arXiv:1501.01685 has a negative answer.

Scope

This identification settles precisely the general "whenever the modulus exists" question. Elliott separately notes that his operator space is not a vector lattice and leaves open the narrower structural variant asking whether the formula must hold when the full space of regular operators is a vector lattice. This packet makes no claim about that variant. The martingale-specific open questions advertised in arXiv:1501.01685 are answered within that same paper and therefore are extraction false positives rather than separate literature results.

Search Evidence

Cheap run indexes were searched for the source arXiv id, title, regular abstract martingales, Riesz–Kantorovich, and modulus of an operator; no duplicate packet was found. A bounded search for the exact question and counterexamples located Elliott's publisher record and decisive article.

References

[1] V. G. Troitsky and F. Xanthos, *Spaces of regular abstract martingales*, arXiv:1501.01685 (2015).

[2] M. Elliott, *The Riesz–Kantorovich formulae*, Positivity **23** (2019), 1245–1259, doi:10.1007/s11117-019-00661-9.